REFINEMENT vs. REFRESH

Compute allocation for closed-loop robot policies

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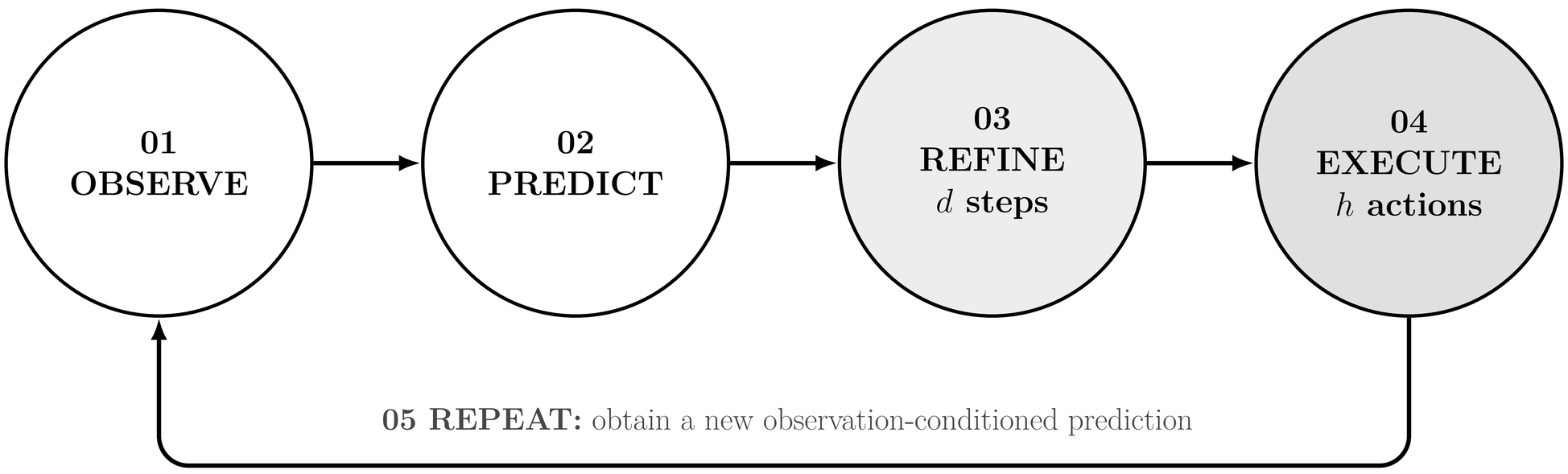
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INFERENCE COMPUTE IS A CONTROL DECISION.

A RECIPE FOR CLOSED-LOOP INFERENCE

Choose how long to refine, how long to execute, and when to observe again.

CHOOSE d AND h UNDER THE AVAILABLE COMPUTE BUDGET



THE HELD-OUT RESULT

FEWER DENOISING STEPS. MORE SUCCESS.

Same two-action execution prefix; eight Goal and Object tasks excluded from configuration selection.



−5.9%
arithmetic per control step with 2 instead of 5 denoising steps

82.1%
of generation arithmetic is independent of denoising count

−60.7%
arithmetic per attempt for $h = 5$ versus $h = 2$, at $d = 2$

HOW WE TESTED IT

12 configurations
4 Spatial tasks
480 discovery attempts

7 frozen configurations
8 held-out tasks
1,120 confirmation attempts

Frozen SmolVLA
Synchronous LIBERO
Paired evaluation

THE IDEA TO REMEMBER

Better robot inference is not only about refining longer. It is also about deciding when to look again.

Evaluate refinement and replanning jointly through closed-loop success and compute.

INTERPRET WITH CARE

Similar observed success for $h = 2$ and $h = 5$ does not establish equivalent quality. Results cover one checkpoint and synchronous simulation.

github.com/amtellezfernandez/refinement-refresh-robot-policies